

# David Sweeney CV

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## Education

- PhD Statistics: University of St Andrews (2025-present)
  - Taking a deeper dive into goose-beaked whale population consequences of disturbance (PCoD)
    - Research supervised by Len Thomas, Enrico Pirotta, Gregory Schorr, and Erin Falcone
- MSc Statistics: University of St Andrews (2024-2025)
  - Graduated with distinction and placed on the Deans' List for academic excellence
  - Using mark-recapture models to understand goose-beaked whale residency in Southern California
    - Research supervised by Len Thomas, Enrico Pirotta, Gregory Schorr, and Erin Falcone
- BS Biology: Calvin College (2015-2019)
  - Graduated with Honors
  - GPA: Within Major = 3.877; Overall = 3.798
  - Au Sable Institute of Environmental Studies
    - Participation in the 2016 Summer Session 2 at the Pacific Rim Campus
    - Courses taken: Marine Biology and Marine Mammals

## Research Employment

- 2024-present: Centre for Research into Ecological and Environmental Modelling ([CREEM](#))
  - Postgraduate Researcher (MSc and PhD)
  - Supervised by Len Thomas and Enrico Pirotta
- 2019-present: Foundation for Marine Ecology and Telemetry Research ([MarEcoTel](#))
  - Full-time from 2019-2024 and then affiliate researcher from 2024-present
- 2018: Duke University Marine Lab
  - Undergraduate Researcher (NSF REU)
  - Supervised by Andrew Read and Nicola Quick
  - Short-finned pilot whale (*Globicephala macrorhynchus*) near-surface behavioral states in relation to previous dive characteristics
- 2017-2019: Calvin University
  - Undergraduate Researcher
  - Supervised by Stacy DeRuiter

- Developing Software Tools for Analysis of Data from Animal-Borne Movement Loggers

## Teaching Employment

- 2025-present: University of St Andrews
  - Tutor and computer practical demonstrator for Statistical Inference
    - Supervised by Len Thomas
  - Computer practical demonstrator for Applied Statistical Modelling using GLMs
    - Supervised by Fergus Chadwick
- 2018-2019: Calvin College
  - Lab Assistant for Human Anatomy
    - Supervised by Ryan Bebej
  - Teaching Assistant for Oceanography
    - Supervised by Ralph Stearley

## Peer-Reviewed Publications

In press:

- DeRuiter, S.L., Falcone, E.A., Coates, S.N., Rone, B.K., **Sweeney, D.A.**, Dolan, K.A., Jarvis, S.M., Andrews, R.D., Watwood, S.L., and Schorr, G.S. (2026). Assessment of goose-beaked whale responses to mid-frequency active sonar using a hierarchical hidden Markov model. *Movement Ecology*. <https://doi.org/10.1186/s40462-026-00647-9>
- Barlow, J., Baird, R. W., Badger, J., Schorr, G. S., Falcone, E. A., Coates, S. N., Tyack, P. L., Read, A. J., Hickmott, L. S., Durban, J. W., Claridge, D., de Soto, N. A., Gonzalez, D. M., Visser, F., Oudejans, M. G., DeRuiter, S., **Sweeney, D. A.**, Rone, B. K., & Watwood, S. L. (2025). A Comparison of Diving Behavior of Goose-Beaked and Dense-Beaked Whales From Tagging Studies in Multiple Ocean Basins. *Marine Mammal Science*. <https://doi.org/10.1111/mms.70070>
- Coates, S. N., **Sweeney, D. A.**, Falcone, E. A., Watwood, S. L., Rone, B. K., DeRuiter, S. L., Barlow, J., Dolan, K. A., Morrissey, R. P., DiMarzio, N. A., Jarvis, S. M., Andrews, R. D., & Schorr, G. S. (2024). Insights into foraging behavior from multi-day sound recording tags on goose-beaked whales (*Ziphius cavirostris*) in the Southern California Bight. *Frontiers in Marine Science*, 11, 1415602. <https://doi.org/10.3389/fmars.2024.1415602>
- Fahlman, A., Schorr, G. S., **Sweeney, D. A.**, Rone, B. K., Coates, S. N., Allen, A. S., López, L. M. M., Jarvis, S. M., & Falcone, E. A. (2025). Modelling the effect of varying metabolic rate and cardiac output on estimated tissue and blood O<sub>2</sub> and CO<sub>2</sub> levels in an extreme deep-diver, the goose-beaked whale (*Ziphius cavirostris*). *Experimental Physiology*. <https://doi.org/10.1113/EP093021>

- Rone, B. K., **Sweeney, D. A.**, Falcone, E. A., Watwood, S. L., & Schorr, G. S. (2022). Movements and diving behavior of Risso's dolphins in the Southern California Bight. *Frontiers in Marine Science*, 9, 873548. <https://doi.org/10.3389/fmars.2022.873548>
- Sweeney, **D. A.**, DeRuiter, S. L., McNamara-Oh, Y. J., Marques, T. A., Arranz, P., & Calambokidis, J. (2019). Automated peak detection method for behavioral event identification: Detecting *Balaenoptera musculus* and *Grampus griseus* feeding attempts. *Animal Biotelemetry*, 7(1), 7. <https://doi.org/10.1186/s40317-019-0169-3>
- **Sweeney, D. A.**, Schorr, G. S., Falcone, E. A., Rone, B. K., Andrews, R. D., Coates, S. N., Watwood, S. L., DeRuiter, S. L., Johnson, M. P., & Moretti, D. J. (2022). Cuvier's beaked whale foraging dives identified via machine learning using depth and triaxial acceleration. *Marine Ecology Progress Series*, 692, 195–208. <https://doi.org/10.3354/meps14068>

In review:

In prep:

- Movements and diving behavior of goose-beaked whales (*Ziphius cavirostris*) at Isla Guadalupe, México (**co-author**)
- Predicting blubber thickness in large whales: an analysis to inform satellite tag development (**co-author**)

## Funding Acquisition

- Taking a deeper dive into goose-beaked whale population consequences of disturbance (PCoD)
  - Funded by the Office of Naval Research (ONR) from 2024 – 2029 for a total of \$629,508 (USD)
  - PIs: Gregory Schorr and Len Thomas
  - Co-investigators: **David Sweeney**, Enrico Pirotta, Erin Falcone, Stephanie Watwood, Stacy DeRuiter, Gustavo Cárdenas-Hinojosa
  - Project focus: Quantifying population consequences of disturbance (PCoD) for goose-beaked whales by integrating long-term photo-ID, tagging, and drone-based data with advanced statistical and ecological modelling to assess impacts of naval training activities in the Southern California Bight.

## Posters and Oral Presentations

Research presentation:

- **Sweeney, D. A.**, Thomas, L., Schorr, G., Pirotta, E., Falcone, E., Keene, E., Rone, B., and DiMarzio, N. (2026). Data integration within a continuous-time open-population mark-recapture model improves monitoring of resident goose-beaked whales by accounting for the

transience effect [oral presentation]. *International Statistical Ecology Conference*. Mérida, México: 10-15 January 2027.

- **Sweeney, D. A.**, Thomas, L., Schorr, G., Pirotta, E., Falcone, E., Keene, E., Rone, B., and DiMarzio, N. (2026). New Mark-Recapture Model Framework Accounts for Transients and Enhances our Understanding of Goose-Beaked Whale (*Ziphius cavirostris*) Residency on a US Navy Training Range in Southern California [short talk]. *26<sup>th</sup> Biennial Conference of the Society for Marine Mammalogy*. San Juan, Puerto Rico: 19-23 October 2026.
- **Sweeney, D. A.**, Schorr, G., Thomas, L., Falcone, E., and Pirotta, E. (2026). Using mark-recapture models to understand goose-beaked whale (*Ziphius cavirostris*) residency on a US Navy training range in Southern California [short talk]. *UK & Ireland Regional Student Chapter for the Society of Marine Mammalogy Conference 2026*. Plymouth, UK: 13-16 January 2026
- DeRuiter, S., Johnson, M., Swift, R., Harris, C., Marques, T., McNamara-Oh, Y. J., and **Sweeney, D.** (2019). Tools and Training to Enhance Capacity for Analyzing High-Resolution Tag Data [co-presented]. *ONR Marine Mammal and Biology Program Review 2019*. Alexandria, VA, USA: 23 April 2019.
- **Sweeney, D. A.** (2019). Short-finned pilot whale (*Globicephala macrorhynchus*) near-surface behavioral states in relation to previous dive characteristics [seminar]. *Calvin College Department of Biology*. Grand Rapids, MI, USA: 1 February 2019.
- **Sweeney, D. A.** (2017). Event Detection [workshop]. *Analysis of Data from High-Resolution Animal-Borne Tags*. University of St Andrews, Scotland: 7-9 August 2017.

#### Research poster:

- **Sweeney, D. A.**, Schorr, G., Thomas, L., Falcone, E., and Pirotta, E. (2026). Using mark-recapture models to understand goose-beaked whale (*Ziphius cavirostris*) residency on a US Navy training range in Southern California. *European Cetacean Society 37<sup>th</sup> Annual Conference*. Dundee, Scotland: 22-24 April 2026
- **Sweeney, D.**, Falcone, E., Watwood, S., DeRuiter, S., Schorr, G. (2022) Cuvier's Beaked Whale Behavioral Responses Persist After Conclusion of Some Navy Sonar Exposures. *24<sup>th</sup> Biennial Conference of the Society for Marine Mammalogy*. West Palm Beach, Florida, USA: 1-5 August 2022
- **Sweeney, D. A.** (2018). Short-finned pilot whale (*Globicephala macrorhynchus*) near-surface behavioral states in relation to previous dive characteristics. *West Michigan Regional Undergraduate Science Research Conference*. Grand Rapids, MI, USA: 10 November 2018.
- **Sweeney, D. A.** (2018). Short-finned pilot whale (*Globicephala macrorhynchus*) near-surface behavioral states in relation to previous dive characteristics. *Calvin Science Division Summer Research Poster Fair*. Grand Rapids, MI, USA: 19 October 2018.
- **Sweeney, D. A.** and Oh, Y. J. (2017). Software Tools for Analysis of Data from High-Resolution Animal-Borne Tags. *West Michigan Regional Undergraduate Science Research Conference*. Grand Rapids, MI, USA: 4 November 2017.

- **Sweeney, D. A.** and Oh, Y. J. (2017). Software Tools for Analysis of Data from High-Resolution Animal-Borne Tags. *Calvin Science Division Summer Research Poster Fair*. Grand Rapids, MI, USA: 20 October 2017.

#### Other:

- **Sweeney, D. A.** (2024) Encountering God in Creation: Spiritual Growth Through Scientific Understanding [presentation]. *Central Kitsap Presbyterian Church's Conference on Science and the Christian Life: The Word and the World*. Bremerton, WA, USA: 8 June 2024
- **Sweeney, D. A.** (2024) Marine Ecology and Telemetry Research [outreach presentation]. *South Kitsap High School AP Biology*. Port Orchard, WA, USA: 6 June 2024
- **Sweeney, D. A.** (2023) Marine Ecology and Telemetry Research [outreach presentation]. *South Kitsap High School AP Biology*. Port Orchard, WA, USA: 15 June 2023
- **Sweeney, D. A.** (2018). Insights into Modern Marine Mammal Research [guest lecture]. *Calvin College Department of Geology, Geography, and Environmental Studies*. Grand Rapids, MI, USA: 5 December 2018.

## Software Development and Management

- [MarEcoTel](#)
  - GitHub organization developer
    - R and Matlab code development
  - [ZcTransCMR](#)
    - Continuous-time open-population mark-recapture modeling
  - [WCtagproc](#)
    - Tools for processing Wildlife Computers satellite tag data
  - [UASMorpho](#)
    - Code to collect, measure, process, and analyze UAS photogrammetry data
- [animaltags](#)
  - Software for data from high-resolution movement tags (<https://animaltags.org/>)
  - GitHub organization developer
    - R, Matlab, and Octave code development

## Field Work Experience

- Southern California Offshore Range (SCORE); Southern California Anti-Submarine Warfare Range (SOAR); Point Mugu Sea Range
  - Vessels: Phoenix and Physalus (2019-2024)
  - Duties: Photo-ID, Database entry, RHIB driving (including for LIMPET tagging and biopsies)

- Species: Cuvier's beaked whales, Fin whales, Bryde's whales, Delphinus spp., Humpback whales
- Guadalupe Island, Mexico
  - Vessel: Azteca (2021-2024)
  - Duties: Drone photogrammetry, Photo-ID, Database entry
  - Species: Cuvier's beaked whales, Bottlenose dolphins, Humpback whales
- Strait of Juan de Fuca and outer coast of Washington, USA
  - Vessel: Borealis (2021-2023)
  - Duties: Photo-ID, Database entry, RHIB driving
  - Species: Killer whales, Humpback whales, Dall's and Harbor porpoises
- Hood Canal, WA
  - Vessel: On Porpoise (2022-2023)
  - Duties: Harbor porpoise survey visual observer, Pontoon boat driving, Database entry
  - Species: Harbor porpoise, Harbor seal, California sea lion, Steller sea lion
- UAS Experience:
  - Remote Pilot Certified (United States FAA #4555354)



## Scholarships/Awards

- Runner-Up for Best Short Talk (UK & Ireland Regional Student Chapter for the Society of Marine Mammalogy Conference 2026)
- Handsel Scholarship (University of St Andrews)
- 2019 Ten Broek Excellence in Biology Research Award (Calvin College)
- Tuition Waiver-CCCU TWEP (Calvin College)
- Honors Fellows Scholarship (Calvin College)
- Presidential Scholarship (Calvin College)
- Au Sable Fellowship (Au Sable Institute of Environmental Studies)
- Swierenga, John R. & Marie A. Brass Scholarship (Calvin College)
- Friends of Music Scholarship (Calvin College)

## **Peer-Reviews [number of papers] and Conference Abstract Reviews**

- Ecological Solutions and Evidence (co-reviewer) [1]
- 26<sup>th</sup> Biennial Conference on the Biology of Marine Mammals
- Marine Biodiversity Records [1]
- Biology Letters (co-reviewer) [1]

## **Non-Research Work and Notable Volunteer Experiences**

- Administrator of Cornerstone St Andrews church (2025-present)
- Co-founder of Central Kitsap Presbyterian Church's Community Garden (2021-2024)
- Assistant Youth Group Director
  - Silverdale Lutheran Church (2019-2021)
  - Central Kitsap Presbyterian Church (2021-2024)
- Assistant Caretaker of Olympic Lutherhaven in Bremerton, WA (2020-present)
- Knollcrest Dining Hall Team Member at Calvin College (Spring 2019)
- Lifeguard at Six Flags Hurricane Harbor in Gurnee, IL (2013-2016)
  - Leadership Team Member (2015-2016)